Nexora Build

Design, Estimate, Build and Manage

PROJECT REFERENCE MANUAL | Verified repository facts separated from previously discussed scope and unverified versions.

Source: https://github.com/Nishanth091123/Nexora-Build

01 | Business problem and end goal

1. Construction design, estimation, site tracking and project records are often fragmented.

2. End goal: Organized project workspaces; Design/estimate records. Use outputs to guide decisions, not to replace expert review.

02 | Inputs and data dictionary

1. Project and user details

2. Plans, measurements and project tasks

3. Site records, photos and estimates

4. Settings and backup inputs

03 | End-to-end processing workflow

1. Create/authenticate project users

2. Prepare design and estimate records

3. Track construction-site activities

4. Manage records and workflows

5. Back up, restore and report

04 | Outputs and how to interpret them

1. Organized project workspaces

2. Design/estimate records

3. Site and task status

4. Backup and recovery records

05 | Important features and why they matter

1. Input measurement and plan accuracy
2. User permissions and auditability
3. Cost/quantity assumptions
4. Data backup and restore

06 | Calculation, decision logic and worked interpretation

1. Repository describes Design -> Estimate -> Build -> Manage.
2. Do not label a feature as machine learning unless an actual model/inference pipeline is verified.
3. Access permissions and recoverability are central to multi-user construction workflows.

07 | Technology stack and version register

1. FastAPI == 0.115.12
2. Uvicorn == 0.34.2
3. SQLAlchemy == 2.0.40
4. psycopg == 3.2.9
5. openpyxl == 3.1.5; reportlab == 4.4.1
6. pypdf == 5.9.0; bcrypt == 4.3.0
7. Frontend versions not verified

Version control rule: '==' is an exact pin; '>=' or '<' is a supported range; an unpinned requirement is not evidence of the installed version. Capture `python --version`, `pip freeze`, database server version, and deployment commit SHA for a reproducible release.

08 | Common confusions, debugging and prevention

1. Keep .env and credentials out of GitHub.

2. Test restore before relying on backups.

3. Validate role access and cross-project data isolation.

4. Check backend/frontend compatibility after upgrades.

09 | Interview preparation questions

1. How does Nexora differ from Civora?

2. How are user roles secured?

3. How are backups restored?

4. Which features actually use AI?

How to answer in an interview

Explain: (1) business problem, (2) input fields, (3) actual processing or model, (4) how the result is calculated, (5) how a user acts on the result, (6) limitations. Distinguish code-confirmed facts from plans.

10 | Maintenance checklist

[ ] Record exact environment and commit SHA

[ ] Keep example input and expected output

[ ] Test boundary and missing-value cases

[ ] Confirm production credentials and backup/restore

[ ] Retest PDF/export, login, API and database flows

[ ] Review the documentation after each release

Source confidence and scope

Repository-verified facts are based on README, requirements or targeted code inspected for this edition. Project-specific figures from prior discussion and planned features are labeled accordingly. This document is a maintained technical reference, not a full line-by-line code audit.